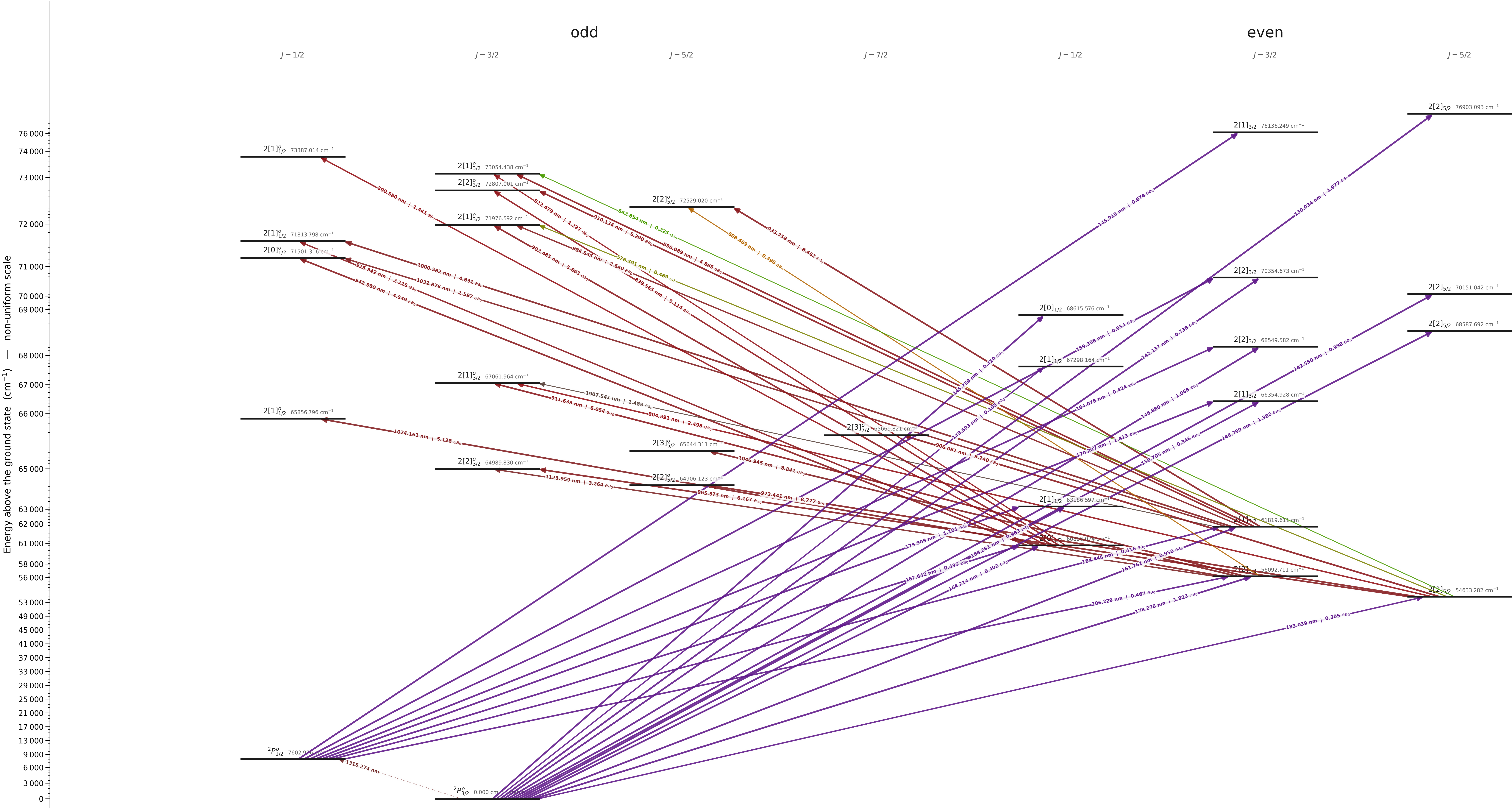
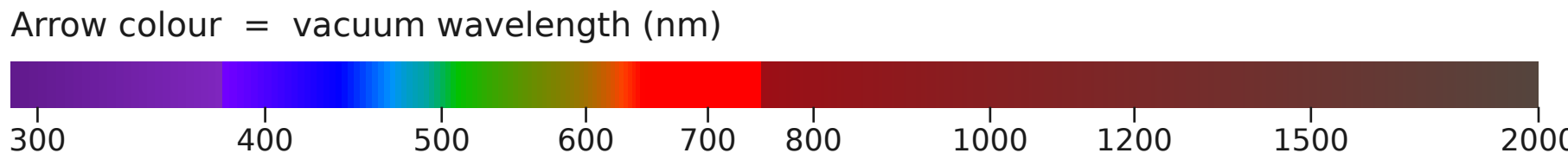


# Iodine

I (I I). Strongest classified transitions below 2 μm (NIST): 29 levels, 46 lines.

I<sup>+</sup> ionisation limit 84294.90 cm<sup>-1</sup> = 10.45124 eV (118.631 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element  $[J||e\mathbf{r}||J']$  in  $e a_0$   
Matrix elements are measured or NIST values; \* = high-accuracy theory, ≈ = model calculation.  
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation,  $n = 70$ .  
Level caption: NIST energy and measured lifetime (where one exists).



## Strongest transitions

| Transition                  | λ vac (nm) | λ air (nm) | ν (THz)   | $[J  e\mathbf{r}  J']$ ( $e a_0$ ) | A (s <sup>-1</sup> ) | Γ/2π up. (MHz) | branch (%) | Use |
|-----------------------------|------------|------------|-----------|------------------------------------|----------------------|----------------|------------|-----|
| $2P_{3/2}^o - 2[2]_{5/2}$   | 130.0338   | -          | 2305.4967 | 1.977                              | 6.000e+08            | -              | -          | -   |
| $2P_{3/2}^o - 2[2]_{3/2}$   | 142.1370   | -          | 2109.1800 | 0.738                              | 9.600e+07            | -              | -          | -   |
| $2P_{3/2}^o - 2[2]_{5/2}$   | 142.5496   | -          | 2103.0753 | 0.998                              | 1.160e+08            | -              | -          | -   |
| $2P_{3/2}^o - 2[0]_{1/2}$   | 145.7395   | -          | 2057.0432 | 0.410                              | 5.500e+07            | -              | -          | -   |
| $2P_{3/2}^o - 2[2]_{5/2}$   | 145.7988   | -          | 2056.2073 | 1.382                              | 2.080e+08            | -              | -          | -   |
| $2P_{3/2}^o - 2[2]_{3/2}$   | 145.8798   | -          | 2055.0648 | 1.068                              | 1.860e+08            | -              | -          | -   |
| $2P_{1/2}^o - 2[1]_{3/2}$   | 145.9145   | -          | 2054.5758 | 0.674                              | 7.400e+07            | -              | -          | -   |
| $2P_{3/2}^o - 2[1]_{1/2}$   | 158.2614   | -          | 1894.2865 | 0.983                              | 2.470e+08            | -              | -          | -   |
| $2P_{1/2}^o - 2[2]_{3/2}$   | 159.3582   | -          | 1881.2485 | 0.954                              | 1.140e+08            | -              | -          | -   |
| $2P_{3/2}^o - 2[1]_{3/2}$   | 161.7610   | -          | 1853.3053 | 0.950                              | 1.080e+08            | -              | -          | -   |
| $2P_{3/2}^o - 2[0]_{1/2}$   | 164.2142   | -          | 1825.6184 | 0.402                              | 3.700e+07            | -              | -          | -   |
| $2P_{1/2}^o - 2[1]_{3/2}$   | 170.2071   | -          | 1761.3392 | 1.413                              | 2.050e+08            | -              | -          | -   |
| $2P_{3/2}^o - 2[2]_{3/2}$   | 178.2763   | -          | 1681.6172 | 1.823                              | 2.970e+08            | -              | -          | -   |
| $2P_{1/2}^o - 2[1]_{1/2}$   | 179.9091   | -          | 1666.3550 | 1.101                              | 2.110e+08            | -              | -          | -   |
| $2[2]_{5/2} - 2[3]_{7/2}^o$ | 906.0811   | 905.8325   | 330.8671  | 9.740                              | 3.230e+07            | -              | -          | -   |
| $2[1]_{3/2} - 2[2]_{5/2}^o$ | 933.7584   | 933.5022   | 321.0600  | 8.462                              | 2.970e+07            | -              | -          | -   |

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.  
Γ/2π: natural linewidth of the upper level, 1/(2πτ).